

Smartphone Ownership, Age of Smartphone Acquisition, and Health

Outcomes in Early Adolescence

Supplemental Material

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Supplemental eMethods 1: Depression, obesity, and sleep outcomes

Depression (lifetime [past or current] diagnosis of either Major Depressive Disorder, Persistent Depressive Disorder (Dysthymia), or Unspecified Depressive Disorder) was based on the Kiddie Structured Assessment for Affective Disorders and Schizophrenia for DSM-5 (KSADS-5),¹ which was conducted at the 2-year and 4-year follow-up assessments. In Analysis 3, as the mood disorder module of the K-SADS was not administered and a depression diagnosis could not be determined, we used a binary outcome variable based on the self-reported Brief Problem Monitor for youth (BPM) T-score ≥ 65 , which meets clinical threshold level of psychopathology symptom burden.² The BPM is a validated tool composed of questions related to internalizing, externalizing, and attention symptoms.³

Obesity was determined using body mass index (BMI, kg/m^2 , continuous measure) based on height and weight measurements. According to CDC definitions,⁴ we considered obesity (binary measure) for participants over the sex- and age-specific 95th BMI percentile. We excluded children with extreme BMI z scores (<-4 or >8).^{5,6}

Insufficient sleep was determined based on the recommendation by the American Academy of Sleep Medicine, considered for participants with <9 hours of night sleep.⁷ Hours of sleep were determined based on the Munich Chronotype Questionnaire, a self-report measure that captures hours of sleep (continuous measure representing average sleep duration across the whole week including school days and free days) based on youth-reported sleep onset and end.⁸ We removed outliers above and below mean $\pm 4\text{SD}$.

Supplemental eMethods 2: Experimental design

Analytic code describing all analyses is available at

https://github.com/barzilab1/ABCD_smartphone_health.

In Analysis 1, we tested the association of smartphone ownership (exposure 1) with health outcomes (depression, obesity, insufficient sleep) at the 2-year follow-up assessment (N=10,588). In Analysis 2, we tested association of age of first smartphone acquisition (exposure 2) with health outcomes (depression,

obesity, insufficient sleep) at the 2-year follow-up assessment among youth who owned a smartphone by this timepoint (n=6,739). In Analysis 3, we included only participants who did not own a smartphone at the 2-year follow-up (n=3,486). We then examined whether acquiring a smartphone by the 3-year follow-up (exposure 3) was associated with health outcomes measured at that same 3-year follow-up.

Supplemental eMethods 3: Statistical models and covariates

Analyses were conducted using the R statistical software version 4.5.0. For each of the health measures, we estimated two mixed models adding covariates in a stepwise manner. First, we included age and sex assigned at birth (Model 1). Thereafter, in model 2, we added demographic and socioeconomic factors, which included household income, parent years of education (highest of the parents), and Black race and Hispanic ethnicity (race and ethnicity were reported by the parent). Model 2 also adjusted for ownership of other devices (tablets, laptops, iPods, and smartwatches). Additional covariates included pubertal status at time of assessment (self-reported, measured on a 1-5 Likert scale from prepubertal to post-pubertal), and parental monitoring as potential confounding factors of pubertal and parenting effects on participant's health outcomes. Parental monitoring was assessed as the mean score of five items from the Parental Monitoring scale:^{9,10} *“How often do your parents/guardians know where you are?”*, *“How often do your parents know who you are with when you are not at school and away from home?”*, *“If you are at home when your parents or guardians are not, how often do you know how to get in touch with them?”*, *“How often do you talk to your parent or guardian about your plans for the upcoming day, such as your plans about what will happen at school or what you are going to do with friends?”*, and *“In an average week, how many times do you and your parents/guardians eat dinner together?”* Youth responses for each item were reported on a five-point Likert scale ranging from 1 (*never*) to 5 (*always or almost always*). The mixed models testing associations of acquisition of smartphone in the past year with health measures after a year (at the 3-year follow-up assessment, Analysis 3) also adjusted for the health measure being tested at the 2-year assessment.

Supplemental eMethods 4: Estimation of Problematic Smartphone Use in ABCD Study

There is no consensus in research on how to quantify problematic smartphone use.¹¹ Here we estimated Problematic Smartphone Use with the Mobile Phone Involvement Questionnaire (MPIQ),¹² which was administered to participants who reported having mobile phones in ABCD Study, and was previously used to assess problematic screen use in ABCD Study.¹³ The MPIQ probes behavioral domains related to problematic smartphone use and requires participants to rate their agreement with the below eight statements on a Likert-type scale:

1. I interrupt whatever else I am doing when I am contacted on my phone
2. I often use my mobile phone for no particular reason
3. I feel connected to others when I use my mobile phone
4. Arguments have arisen with others because of my mobile phone use
5. I lose track of how much I am using my mobile phone
6. I often think about my phone when I am not using it
7. I have been unable to reduce my mobile phone use
8. The thought of being without my mobile phone makes me feel distressed

Likert scale ranged from 1-7 as follows: 1=Strongly Disagree; 2=Disagree; 3=Somewhat Disagree; 4=Neither Agree nor Disagree; 5=Somewhat Agree; 6=Agree; 7=Strongly Agree.

For the current work, we created a variable indicating Problematic Smartphone Use by binarizing the mean score using the cutoff of 5, whereby a mean score of above 5 (i.e., somewhat agreeing with the statements) may represent participants with problematic smartphone use.

Supplemental eMethod 5: Assumptions and sensitivity analysis

Since our analysis of observational data depends on an assumption of no unobserved confounding between treatment and control conditions, we conduct a sensitivity analysis to quantify the degree of unobserved confounding required to explain our results in the absence of a treatment effect. As the main

analyses are conducted in a mixed model framework, we assess the degree of unobserved confounding necessary to reverse our findings by computing E-values for analyses of binary outcomes, which allows us to keep with the mixed model analytic framework. The E-value represents the association an unmeasured confounder would have to exhibit (on the risk-ratio scale) with both treatment and outcome to explain the observed results if the true treatment effect is zero. We conducted E-value analysis for the logistic models that included the largest number of covariates (Model 2).

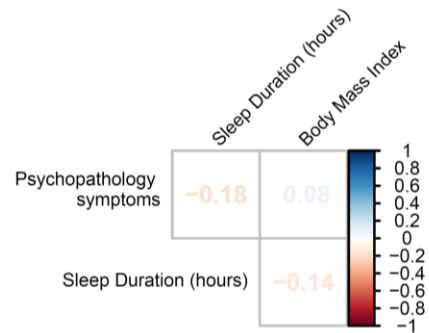
Supplemental eMethod references

1. Townsend L, Kobak K, Kearney C, et al. Development of Three Web-Based Computerized Versions of the Kiddie Schedule for Affective Disorders and Schizophrenia Child Psychiatric Diagnostic Interview: Preliminary Validity Data. *J Am Acad Child Adolesc Psychiatry*. 2020;59(2):309-325. doi:10.1016/j.jaac.2019.05.009
2. Achenbach TM, McConaughy SH, Ivanova MY, Rescorla LA. Manual for the ASEBA brief problem monitor (BPM). In: *ASEBA*. ; 2011:1-33.
3. Piper BJ, Gray HM, Raber J, Birkett MA. Reliability and validity of brief problem monitor, an abbreviated form of the child behavior checklist. *Psychiatry Clin Neurosci*. 2014;68(10):759-767. doi:10.1111/PCN.12188/SUPINFO
4. Ogden CL, Flegal KM. Changes in terminology for childhood overweight and obesity. *Natl Health Stat Report*. 2010;(25):1-5. Accessed July 12, 2021. <https://europepmc.org/article/med/20939253>
5. Kuczmarski RJ, Ogden CL, Guo SS, et al. 2000 CDC Growth Charts for the United States: methods and development. *Vital Health Stat 11*. 2002;(246):1-190.
6. Kaltenhauser S, Weber CF, Lin H, et al. Association of Body Mass Index and Waist Circumference With Imaging Metrics of Brain Integrity and Functional Connectivity in Children Aged 9 to 10 Years in the US, 2016-2018. *JAMA Netw Open*. 2023;6(5):e2314193-e2314193. doi:10.1001/JAMANETWORKOPEN.2023.14193
7. Paruthi S, Brooks LJ, D'Ambrosio C, et al. Consensus Statement of the American Academy of Sleep Medicine on the Recommended Amount of Sleep for Healthy Children: Methodology and Discussion. *Journal of Clinical Sleep Medicine*. 2016;12(11):1549-1561. doi:10.5664/JCSM.6288
8. Yang FN, Picchioni D, Duyn JH. Effects of sleep-corrected social jetlag on measures of mental health, cognitive ability, and brain functional connectivity in early adolescence. *Sleep*. 2023;46(12):1-9. doi:10.1093/SLEEP/ZSAD259
9. Karoly HC, Callahan T, Schmiede SJ, Feldstein Ewing SW. Evaluating the Hispanic Paradox in the Context of Adolescent Risky Sexual Behavior: The Role of Parent Monitoring. *J Pediatr Psychol*. 2016;41(4):429-440. doi:10.1093/JPEPSY/JSV039
10. Zucker RA, Gonzalez R, Feldstein Ewing SW, et al. Assessment of culture and environment in the Adolescent Brain and Cognitive Development Study: Rationale, description of measures, and early data. *Dev Cogn Neurosci*. 2018;32:107-120. doi:10.1016/J.DCN.2018.03.004
11. Harris B, Regan T, Schueler J, Fields SA. Problematic Mobile Phone and Smartphone Use Scales: A Systematic Review. *Front Psychol*. 2020;11:672. doi:10.3389/fpsyg.2020.00672
12. Walsh SP, White KM, Young RM. Needing to connect: The effect of self and others on young people's involvement with their mobile phones. *Aust J Psychol*. 2010;62(4):194-203. doi:10.1080/00049530903567229

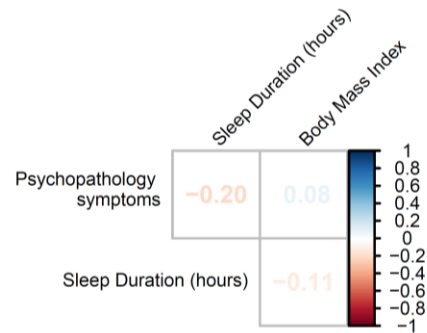
13. Chu J, Ganson KT, Testa A, et al. Screen time, problematic screen use, and eating disorder symptoms among early adolescents: findings from the Adolescent Brain Cognitive Development (ABCD) Study. *Eating and Weight Disorders*. 2024;29(1):1-10. doi:10.1007/S40519-024-01685-1/TABLES/3

Supplemental Figure S1: Correlation among health outcomes

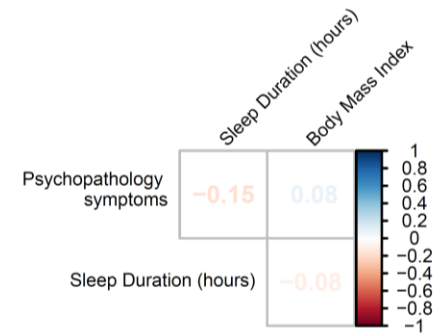
Correlation at 2-year follow-up



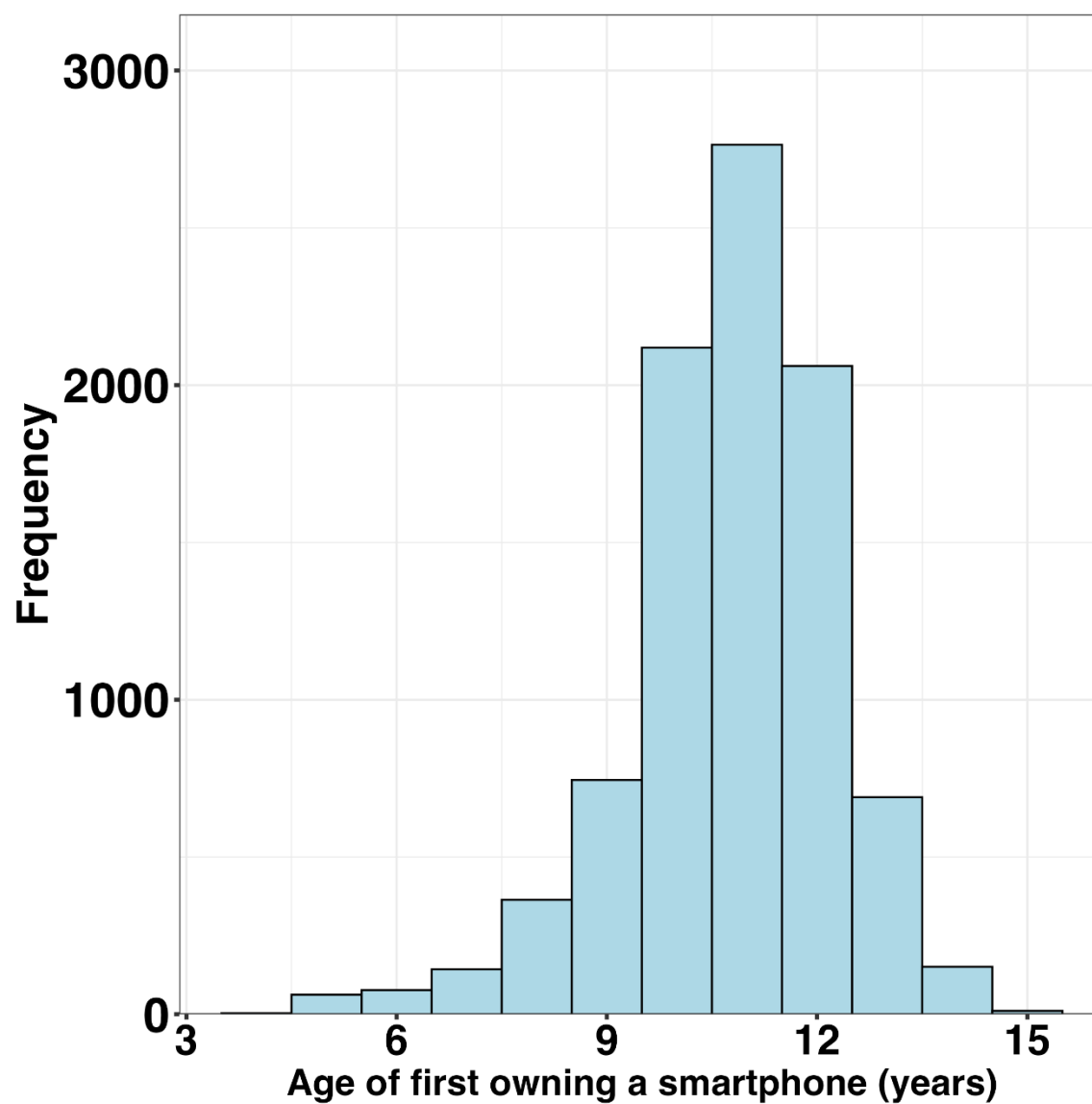
Correlation at 3-year follow-up



Correlation at 4-year follow-up



Supplemental Figure S2: Distribution of age-of-onset for smartphone ownership as reported across three assessment waves (2-year, 3-year and 4-year follow-up).



Median age 11, range 4-15, interquartile range 10-12.